

“AI Computers get **expensive very fast**. You can end up spending **\$1000**”

Over the past few years, a lot has been happening in AI in both hardware and software. We have seen new algorithms, new processing techniques, and new AI chips. **Jim Keller, CTO and president of Tenstorrent**, an AI startup, sheds light on these cutting-edge technologies in interview with *EFY technologies*.

Jim Keller, CTO and president, Tenstorrent »



Q. What motivated you to get involved in a startup after multiple roles in corporate companies? How has been your experience at Tenstorrent?

A. When I was at Tesla, a whole bunch of AI startups were coming and trying to pitch Tesla on their AI stuff. Then I went to Intel, which was one of the challenges of a lifetime—a team of 10,000 people! When I left Intel, I thought about starting a new company from scratch, but the AI revolution had already started. So, I joined the company (Tenstorrent).

I was also their first investor. Ljubisa Bajic started the company and he called me and said, “Hey, we have this new idea to do an AI processor that’s different and better” and I gave him what’s called the angel investment. We thought we could bring forth

something unique by combining a really great AI processor and a GPU together in a way no other AI startup was doing.

But for certain reasons, I also took over the business side—operations, HR, and legal stuff. And I enjoyed that kind of work as well. In a small company, you get to do these things from scratch. You get exposed to the details of everything. It’s very refreshing. It’s a big contrast from a big company.

Q. How are AI programs different from traditional ones?

A. So, first of all, AI programs are very different from regular programs. In those programs, there’s a serial or sequential flow. You have some branches back and forth. You may have many processors, but each one is running threads. It’s easy for humans to

read it because humans write the code.

AI programs say something like this, “Take some information, represent it like an image or a very long string of sentences, and then multiply it by a very large array of numbers, and then do that a thousand times.” As you multiply by the numbers, you’re finding out the associations of bad information with previously stored information in some subtle but distributed way. It goes through two steps; you train the model (the set of operations is called a model) and you have an expected result.

Say, I want to complete this sentence, or I want to identify an object in a picture. When you start the model, it has no information in it. So, as you train the model, it starts to understand the relationship between the inputs and the stored information. And that’s the AI revolution.